

KARUSH SURI

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AUTODIDACTIC LEARNING

Reading

Topology, James Munkres, all chapters	2026
Topics in Algebra, I.N. Herstein, all chapters	2025
Optimization Algorithms on Matrix Manifolds, P. Absil, chapters 2-7	2021
Convex Optimization, Stephen Boyd, chapters 6-11	2021
Neuro-dynamic Programming, Dimitri Bertsekas and John Tsitsiklis, chapters 1-6	2021
Abstract Dynamic Programming, Dimitri Bertsekas, chapters 1-2	2020
Machine Learning: A Probabilistic Perspective, Kevin Murphy, chapters 20-24	2020
Pattern Recognition and Machine Learning, Christopher Bishop, chapters 8-12	2019

Writing

Paper Summaries	(link)
Research Work	(link)

Code Implementations

Toy Examples (JAX and PyTorch)	(link)
Hierarchical DQN (JAX)	(link)
Proximal Policy Optimization (JAX)	(link)
Conservative Q Learning (JAX)	(link)
Discrete Gumbel Samplers (JAX)	(link)
Eligibility Traces with Neural Networks (PyTorch)	(link)
Evolution Strategies (PyTorch)	(link)
Lagrangian Algorithms and Vector Products (Autograd)	(link)

EDUCATION

University of Toronto	2025 - 2028 (Expected)
<i>PhD, Electrical & Computer Engineering</i>	<i>Toronto, Canada</i>
Thesis: Tail End Gumbel Post-Training	
GPA: 3.8 / 4 (minor in mathematics)	
University of Toronto	2019 - 2021
<i>MASc, Electrical & Computer Engineering</i>	<i>Toronto, Canada</i>
Thesis: Deep Hierarchical Reinforcement Learning	
GPA: 4/4	
Amity University	2015 - 2019
<i>BTech, Electrical & Computer Engineering</i>	<i>Delhi, India</i>
Thesis: Sign Language Translation from Wearable Sensors (demo)	
GPA: 8.78/10 (rank: 4/142)	

EMPLOYMENT

Valence Labs (Mila)	2023 - 2025
<i>Research Engineer</i>	<i>Montreal, Canada</i>
Advisors: Dr. Emmanuel Bengio	
Projects: (1) Multi-Modal Phenomic Foundational Models, (2) Amortized Inference with GFlowNets	
Google X	2021 - 2023
<i>AI Resident</i>	<i>Mountain View, USA</i>
Advisors: Grace Brentano & Dr. Lam Nguyen	
Project: Undisclosed	

INTERNSHIPS

Lossfunk <i>Research Intern</i> Advisors: Paras Chopra Projects: Understanding Drug Repositioning via Chemical Reasoning	2026 - 2026 Toronto, Canada
Fidelity Investments <i>Machine Learning Intern</i> Advisors: Kyle Burgers Projects: Sparse Transformers for Tabular Quality Control	2025 - 2026 Toronto, Canada
Borealis AI <i>Student Researcher</i> Advisors: Xiao Qi Shi Projects: Deep Hierarchical Reinforcement Learning for Trade Execution	2020 - 2021 Toronto, Canada

PUBLICATIONS

* = equal contribution

<i>“Crosscoder Steering Structures and Stabilizes Multi-Answer Biological Reasoning”</i> <u>Karush Suri</u> , Konstantinos Plataniotis, Yuri Lawryshyn, Paras Chopra	Under Review
<i>“Understanding Judge Calibration in Multi-Turn Debates”</i> <u>Karush Suri</u> , Konstantinos Plataniotis, Yuri Lawryshyn	Under Review
<i>“Understanding and Mitigating Overconfidence in Focus Group Surveys”</i> <u>Karush Suri</u> , Konstantinos Plataniotis, Yuri Lawryshyn	TMLR 2026
<i>“A Cross Modal Knowledge Distillation & Data Augmentation Recipe for Improving Transcriptomics Representations through Morphological Features”</i> Ihab Bendif, Yassir El Mesbahi, Alisandra Kaye Denton, <u>Karush Suri</u> , Kian Kenyon-Dean, Auguste Genovesio, Emmanuel Noutahi	ICML 2025
<i>“Understanding Conditional Computation in Contrastive Phenomic Retrieval”</i> <u>Karush Suri</u> , Puria Moghadam, Frederik Wenkel, Maciej Sypetkowski, Emmanuel Bengio, Emmanuel Noutahi, Dominique Beaini	Technical Report 2025
<i>“How Molecules Impact Cells: Unlocking Contrastive PhenoMolecular Retrieval”</i> Philip Fradkin*, Puria Moghadam*, <u>Karush Suri</u> , Frederik Wenkel, Ali Bashashati, Maciej Sypetkowski, Dominique Beaini	NeurIPS 2024 FM4S @ NeurIPS 2024 (oral) (best paper award)
<i>“On the Scalability of GNNs for Molecular Graphs”</i> Maciej Sypetkowski, Frederik Wenkel, Farimah Poursafaei, Nia Dickson, <u>Karush Suri</u> , Philip Fradkin, Dominique Beaini	NeurIPS 2024
<i>“Surprise Minimizing Multi-Agent Learning with Energy-based Models”</i> <u>Karush Suri</u> , Xiao Qi Shi, Konstantinos Plataniotis, Yuri Lawryshyn	NeurIPS 2022
<i>“Off-Policy Evolutionary Reinforcement Learning with Maximum Mutations”</i> <u>Karush Suri</u>	AAMAS 2022 (oral)
<i>“Continuous Sign Language Recognition from Wearable IMUs using Deep CapsNet and Game Theory”</i> <u>Karush Suri</u> , Rinki Gupta	CEE, Elsevier, 2019
<i>“Transfer Learning for sEMG-based Hand Gestures using Deep Learning in a Master- Slave Architecture”</i> <u>Karush Suri</u> , Rinki Gupta	IEEE IC3I 2018

PENDING PATENTS

<i>“Utilizing contrastive machine learning models to extract joint-space molecular-phenomic embeddings from molecular structures or phenomic images”</i> Dominique Beaini, Frederik Wenkel, <u>Karush Suri</u> ,	2025
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Maciej Sypetkowski, Philip Fradkin, Puria Moghadam
Valence Labs, Application Number: WO2026096183A1.

“*Generating Actions for a Supply Chain Network*” 2024
Lam Nguyen, Grace Brentano, Sze Lee, Karush Suri, Anikait Singh,
Salil Pradhan, David Andre
Google X, Application Number: WO2025090795A1.

“*Large Language Model Interface for Supply Chain Networks*” 2023
Lam Nguyen, Grace Brentano, David Andre, Salil Pradhan, Anikait Singh, Karush Suri
Google X, Application Number: US20240289733A1.

SERVICES

Reviewer, <i>TMLR</i>	2026 - Present
Reviewer, <i>ICML</i>	2024 - 2026
Reviewer, <i>ICLR</i>	2024 - 2026
Reviewer, <i>NeurIPS</i>	2024 - 2026
Teaching Assistant, <i>Digital Image Processing</i> , University of Toronto	2021
Teaching Assistant, <i>Computational Thinking</i> , University of Toronto	2021
Teaching Assistant, <i>Computer Networks</i> , University of Toronto	2020
Teaching Assistant, <i>Computer Organization</i> , University of Toronto	2020

AWARDS

Academic

NSERC Canada Graduate Research Scholarship	2026 - 2029
University of Toronto Doctoral Scholarship	2025 - 2029
Hatch Graduate Scholarship	2025 - 2026
Laura Bassi Scholarship (declined)	2025
Electrical & Computer Engineering Fellowship, University of Toronto	2020 - 2021
Edward S. Rogers Graduate Scholarship, University of Toronto	2019 - 2020
Best in Technical Innovation Award (Class of 2015-2019), Amity University	2019
Most Frugal Innovation Award, Amity University	2018
100% Curriculum Merit Scholarship, Amity University	2015
Young Achievers in Mathematics Award	2015

Industrial

Outstanding Reviewer, NeurIPS, ICML	2023, 2026
Outstanding Intern Buddy, Google X	2022